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f26.207.quiz07::30

In all quizzes and exams, show your work and write with correct notation.

(10pts) 1. If y depends on x according to $x + y + x^3 = y^6 + \sin(y)$, find $\frac{dy}{dx}$

$$1 + \frac{dy}{dx} + 3x^2 = 6y^5 \cdot \frac{dy}{dx} + \cos(y) \cdot \frac{dy}{dx}$$

$$1 + 3x^2 = 6x^5 \frac{dy}{dx} + \cos y \cdot \frac{dy}{dx} - \frac{dy}{dx}$$

$$1 + 3x^2 = (6x^5 + \cos y - 1) \cdot \frac{dy}{dx}$$

$$\frac{1 + 3x^2}{6x^5 + \cos y - 1} = \frac{dy}{dx}$$

(10pts) 2. If $y = \sin(x)^{x^8+10x}$ find $\frac{dy}{dx}$

$$\ln(y) = \ln(\sin x^{x^8+10x})$$

$$\ln(y) = (x^8 + 10x)\ln(\sin x)$$

parens

$$\frac{1}{y} \cdot \frac{dy}{dx} = (x^8 + 10x) \cdot \frac{1}{\sin x} \cdot \cos x + \ln(\sin x) \cdot (8x^7 + 10)$$

$$\frac{dy}{dx} = y\left((x^8 + 10x) \cdot \frac{1}{\sin x} \cdot \cos x + \ln(\sin x) \cdot (8x^7 + 10)\right)$$

$$\frac{dy}{dx} = \sin x^{x^8+10x}\left((x^8 + 10x) \cdot \frac{1}{\sin x} \cdot \cos x + \ln(\sin x) \cdot (8x^7 + 10)\right)$$

parens

(10pts) 3. Find an equation for the tangent line to $\boxed{f(x) = 4\sin(x) + \ln(x)}$ at $x = \frac{7\pi}{6}$. Within your equation, simplify the values of all trigonometric expressions.

$$f'(x) = 4\cos x + \frac{1}{x}$$

$$f'\left(\frac{7\pi}{6}\right) = 4\cos\frac{7\pi}{6} + \frac{1}{\frac{7\pi}{6}} = 4 \cdot \frac{-\sqrt{3}}{2} + \frac{6}{7\pi} = -2\sqrt{3} + \frac{6}{7\pi}$$

$$f\left(\frac{7\pi}{6}\right) = 4\sin\left(\frac{7\pi}{6}\right) + \ln\left(\frac{7\pi}{6}\right) = 4 \cdot \frac{-1}{2} + \ln\left(\frac{7\pi}{6}\right)$$

$$y - \left(-2 + \ln\left(\frac{7\pi}{6}\right)\right) = \left(-2\sqrt{3} + \frac{6}{7\pi}\right)\left(x - \frac{7\pi}{6}\right)$$

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